

Optimization and Characterization of 3D Bioprintable Alginate and Hydroxyapatite Based Biomaterial Ink

Kavita Kumari Thakur¹  · Ramesh Lekurwale¹ · Sangita Bansode¹ · Rajesh Pansare¹

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Abstract 3D bioprinting can be utilised to create complex tissues with fine control over the tissue's creation process. 3D bioprinting technology deposits materials layer by layer to obtain customized geometries for creating functional tissue. The 3D bioprinted scaffold provides cells with an ideal environment to grow and proliferate. Customized constructs can be printed by 3D bioprinting techniques by using different biomaterials and cells. Scaffold properties can be tailored for geometries, porosities as well as mechanical properties. This study reports the synthesis of sodium alginate (Alg)/hydroxyapatite (HA) based bioink which can be printable from an extrusion-based bioprinter and is structurally stable for making bone scaffolds. A hydrogen bond formation between hydroxyl groups and carboxyl groups of alginate with hydroxyl groups of hydroxyapatite assures the stability of HA in the alginate hydrogel system. 1 M calcium chloride (CaCl_2) solution used for the post-printing crosslinking. Prepared bioink were characterized using ESEM (environmental scanning electron microscopy), FTIR (Fourier transform infrared spectroscopy), EDS (Energy dispersive X-ray spectroscopy), DTA (Differential thermal analyzer), and TGA (Thermogravimetric analyzer) tests. Rheological and printability responses were evaluated with a Rheometer. FTIR study confirmed the presence of sodium alginate and HA in the bioink. Different shifts and peaks in C=O and O–H obtained from FTIR indicate the presence of ionic interaction between sodium alginate and HA. ESEM analysis showed a rough morphology in the printed structure. EDS

results confirmed the different elements of sodium alginate and HA composites. The calcium/phosphorus (Ca/P) ratio obtained from EDS is in the range of 1.12–1.68. The obtained range for calcium/phosphorus (Ca/P) ratio from EDS analysis ensured the suitability of the bioprinted scaffold for bone regeneration. TGA/DTA results indicated the thermal behaviour and stability of sodium alginate and HA. The viscosity of all the formulated ink of alginate and hydroxyapatite is in the range of viscosity of Extrusion based bioprinter i.e. 30 mPa/s to $>6 \times 10^7$ mPa/s. Rheological studies confirmed that the obtained biomaterial-based ink is in the acceptable range suitable for extrusion-based bioprinters. It also confirmed the shear-thinning properties in the prepared ink. The addition of HA increased the shape retention properties of bioprinted constructs. The combination of sodium alginate and HA provided a suitable biomaterial ink for printing bone scaffolds which has the desired properties for bone regeneration.

Keywords Bioprinting · Bone scaffold · Hydroxyapatite · Biomaterials · Alginate

Introduction

There are various reasons like congenital abnormalities, trauma, diseases, and infections for the occurrences of bone defects. Bone loss can also occur due to traumatic injuries. Congenital conditions are the main reason for abnormalities development in bone [1]. Bone defects can occur due to diseases like osteoporosis, bone tumours or osteogenesis imperfecta [2]. Such bone defects can be treated by processes like surgical procedures, bone grafting and tissue engineering approaches. However, bone grafting is the most preferred technique for treating bone defects. During bone grafting,

✉ Kavita Kumari Thakur
kavitakumari.t@somaiya.edu

¹ Department of Mechanical Engineering, K.J. Somaiya College of Engineering, Vidyavihar, Somaiya Vidyavihar University, Mumbai 4000 77, Maharashtra, India

a surgical method is performed to repair and regenerate bone. Here, bone grafting aims to provide structural support, improve new bone growth processes and enhance the healing process of bone [3]. Bone grafting is also intended to improve bone healing, promote new bone formation, and give structural support. Biological grafts include techniques such as autografting, allografting, and xenografting, whereas ceramic and metal-based grafts are classified as non-degradable. Autologous bone grafting is the most preferred bone grafting technique due to its osteoconduction, osteoinduction and osteogenesis properties for bone regeneration [4]. Although bone grafting is widely used for bone repair and regeneration, there are significant limitations to this technique, including limited donor availability, donor site morbidity, disease transmission from animals, graft resorption, graft failure, and high cost.

To resolve these issues, additive manufacturing and 3D bioprinting technology are used to provide cost-effective and customized treatment in the biomedical field [5]. Bioprinting, also known as the biological cousin of 3D printing, builds a 3D scaffold by layering biocompatible materials together [6]. Advanced tissue regeneration and regenerative medicine are using Extrusion-based bioprinting. This technique uses a combination of biomaterials and cells for the bioink to print a 3D layer-by-layer structure. Extrusion-based bioprinting is used for a wider range of bioink, like hydrogels, cell-laden polymers and biocompatible materials. The choice of biomaterials and cells is crucial in extrusion-based bioprinting to acquire the structural construction needed to form a tissue scaffold. These scaffolds are printed and then cultured *in vitro* before being used *in vivo* [7].

Extrusion-based printing is good at mimicking bone-like tissue constructs [5]. Hydrogels are the easily printable ink for the extrusion-based bioprinter. Hydrogels are a combination of cross-linked polymer chains and water. It has a unique property for mimicking the natural extracellular matrix which is found in living tissues [8]. 3D Scaffolds with complex features can be bioprinted with hydrogels. In the bioprinting process, hydrogels promote cell growth and viability [9]. In this work, the combination of alginate and hydroxyapatite hydrogel has been used as biomaterial-based ink.

Alginate has the unique physical and chemical properties as a biomaterial. It is a natural polysaccharide which is derived from brown seaweed which consists of guluronic and mannuronic acid [10]. This composition of alginate makes it excellent biocompatible for cell encapsulation for tissue engineering applications. Alginate is suited for biomedical applications due to its superior gelling qualities [11]. Alginate could form a gel-like structure by crosslinking with calcium cation [12]. With the help of this, crosslinking stable structure can be obtained in the bioprinting process. Alginate has the shear thinning properties which helps in

easy extrusion from the nozzle, it also recovers its viscosity after the removal of extrusion pressure and maintains its integrity. These properties of alginate make it a suitable candidate for cell growth and tissue development.

Hydroxyapatite is a calcium phosphate compound. Its chemical composition is like the natural bone. Hydroxyapatite ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) is the main mineral component of bone and it is mostly researched for orthopaedic applications [13]. Hydroxyapatite provides a condition for the extracellular matrix. It also provides hardness and stability to the printed construct. Hydroxyapatite has an osteoconductive nature which helps in bone regeneration by supporting cell adhesion and proliferation [14]. Hydroxyapatite has chemical properties like natural bone, because of which it is mostly used in the creation of bone scaffolds. Hydroxyapatite-based bioink can be precisely 3D bioprinted which helps to create patient-specific implants and scaffolds.

Extrusion-based bioprinting technology is dependent upon the suitable ink formation. If cells are loaded in ink preparation, it is known as bioink. If cells are not loaded while preparation of ink, it is known as biomaterial-based ink. On biomaterial-based ink, cells are seeded on the printed construct [15]. Iglesias et al. [16] also reported the printability and compatibility of alginate, hydroxyapatite-based ink, and their biocompatibility. The combination of alginate and hydroxyapatite bioink is also suggested by Jiang et al. [17].

The present study aims to develop the alginate and hydroxyapatite-based biomaterial ink and conduct the rheology testing, ESEM (environmental scanning electron microscopy), EDS (Energy dispersive X-ray spectroscopy), FTIR (Fourier transform infrared spectroscopy), DTA (Differential thermal analyzer), and TGA (Thermogravimetric analyzer) characterization to show the compatibility of the selected biomaterial ink for bone scaffold. To the author's knowledge, these characterizations were not conducted in the previous studies.

Materials and Methods

The current section describes several biomaterials, the process of preparing ink for the present study, and the characterization of the obtained biomaterial ink.

Materials

Alginic acid sodium salt from brown Algae (guluronic acid/mannuronic acid ratio of 6.13, Mw 2769 kDa, ~15,000 cps), calcium chloride (CaCl_2) (Mw 110.98, 99.99% purity) and Hydroxyapatite (HA) (Mw 1004.6 g/mol, reagent grade purity, micro powder) were obtained from Ssynergy Kolkata,

India. Distilled water was purchased from J. Parekh & company Mumbai, India.

Preparation of Alginate- HA Composite

Various compositions of biomaterial ink were prepared using various combinations of alginate and hydroxyapatite. First, hydroxyapatite suspension of concentrations (0, 8, 16 and 24 w/v %) has been made with the help of deionized water at room temperature wherein the electromagnetic stirrer was used for uniform mixing. Then the alginate (6 w/v %) was dissolved in prepared hydroxyapatite suspension by magnetic stirrer at 45 °C and 750 rpm. 20 ml of water was used for ink preparation. Table 1 represents the different combinations of alginate, hydroxyapatite and calcium chloride selected for this study.

Extrusion-Based Bioprinting of Prepared Biomaterial Ink

The prepared biomaterial ink was printed using an extrusion-based 3D bioprinter (Trivima Basic). The prepared ink was filled in a 3 ml syringe wherein a 21-gauge needle was used for printing. The loaded cartridge was centrifuged for 5 min to remove the trapped air bubbles. For obtaining the optimised pressure, the extrusion pressure was varied between 1.10 and 1.40 bar and at a printing speed of 200 mm/min. The thickness of each layer was 0.2 mm. 3D printed alginate and hydroxyapatite-based biomaterial ink with a dimension of $20 \times 20 \times 0.8$ mm was printed. 1M CaCl_2 was used for crosslinking on the printed

constructs. After printing and cross-linking samples were dried at 60 °C and stored in a Petri dish for further characterization. Figure 1 shows the bioprinted scaffold after printing, crosslinking and drying.

Characterization of the Obtained Biomaterial Ink

Rheological testing was performed in the Advanced Rheology department at the Indian Institute of Technology Bombay (IITB) with MCR 702 Rheometer instrument from Anton Paar. At room temperature, the shear rate from 0.01 to 1000 s^{-1} was varied for performing a steady rate sweep test and with the help of this variation viscosity and the shear stress were measured.

Surface morphology of the printed scaffold was conducted with the help of FEI Quanta 200 scanning electron microscope with 15 kV at IITB, India. Before performing the SEM analysis, samples were coated with titanium coating.

EDS mapping was also conducted to characterize the scaffold's composition, especially in terms of the Ca(Calcium)/P(Phosphorus) ratio. Ca/P ratio is important for showing the bone regeneration capability of a bioprinted scaffold. Environmental Scanning Electron Microscope (ESEM) analysis of the bioprinted scaffold was conducted with the help of FEI Quanta 200 scanning electron microscope with 15 kV at IITB, India.

Fourier transform infrared spectroscopy (FTIR) analysis is conducted to obtain molecular structure and chemical composition of biomaterials for bone scaffold. FTIR analysis is conducted on 3000 Hyperion Microscope with Vertex 80 FTIR System, Bruker, Germany at IITB.

Thermogravimetric Analysis (TGA) and Differential Thermal Analysis (DTA) studies were carried out by diamond TG/DTA, Perkin Elmer, USA at IITB, India. TGA and DTA were investigated on various compositions of ink. Parameters for TGA-DTA analysis are heating the samples at 20 Cel/min , in a nitrogen atmosphere between 25 and $600 \text{ }^\circ\text{C}$ [18]

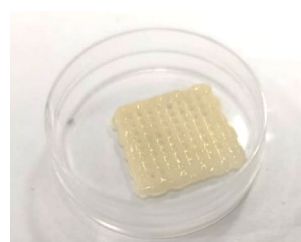
Table 1 Different combinations of biomaterials selected for this study

Biomaterial Ink	Alginate (wt%)	Hydroxyapatite (wt%)	CaCl_2 Molar (M)
Pure 6 wt%	6	0	1
6 wt% + 8 wt%	6	8	1
6 wt% + 16 wt%	6	16	1
6 wt% + 24 wt%	6	24	1

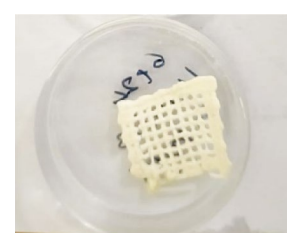
Fig. 1 Bioprinting of the scaffold



(a) Bioprinter



(b) Bioprinted scaffold



(c) Dried scaffold

Results and Discussion

The current section describes the obtained results during the current study. The results are further discussed to understand its importance.

Rheological Testing

To have an idea about the flow of the ink and to obtain the ideal ink for bioprinting, it is very important to understand the rheological behaviour of the preparation [19]. Rheological testing provides information about the viscosity and the shear-thinning behaviour of the prepared ink for optimising the printing parameters for better accuracy as well as resolution. 3D printable inks should have the shear thinning behaviour for the continuous extrusion of the prepared ink. Reduction in viscosity is the result of shear-induced conformation change at a molecular chain. Table 2 represents the values of shear stress and viscosity by varying shear rate for Pure Alg, 6 wt%, 6 wt% Alg + 8 wt% HA, 6 wt% Alg + 16 wt% HA and 6 wt% Alg + 24 wt% HA. Figure 2 displays the pseudo-plastic behaviour of the ink material for different compositions of biomaterial-based ink.

Increasing the shear rate results in a decrease in the viscosity of the ink material displays the pseudoplastic behaviour of the ink material. The ink possesses attributes of a shear thinning fluid, where an increase in shear rate causes a significant reduction in the viscosity of ink material [17]. Increasing hydroxyapatite content increases the viscosity of ink. The viscosity of all the formulated ink of alginate and hydroxyapatite is in the range of viscosity of Extrusion based bioprinter i.e. 30 mPa/s to $> 6 \times 10^7$ mPa/s [20]. Ink with 6 wt% Alg + 8 wt% HA, 6 wt% Alg + 16 wt% HA and 6 wt% Alg + 24 wt% HA shows similar viscoelastic properties. The ink consists of 6 wt% Alg + 24 wt% HA and has maximum viscosity because of which it shows better printability while printing. Ink with 6 wt% Alg + 8 wt% HA and 6 wt% Alg + 16 wt% HA, shows fluid-like behaviour while printing. Similar kinds of rheology studies conducted by the researcher Paxton et al. [21]. Storage Modulus(G') and Loss Modulus(G'') conducted on the prepared biomaterial inks show the elastic behaviour and viscous behaviour of the materials. In Fig. 2c and d, Loss modulus is higher as compared to the shear modulus which represents the viscous behaviour dominates in the formulated composition. The ink consists of 6 wt% Alg + 24 wt% HA and has maximum loss modulus and storage modulus which shows that this composition is highly viscous and is suitable for achieving higher printing resolution.

FTIR Analysis

FTIR analysis on the printed hydrogen shows the common group in their constituent. It provides information on the chemical composition and molecular structure of the biomaterials used in this composition. FTIR helps in identifying the suitability of the biomaterial-based ink for cell attachment and tissue growth. Adhikari et al. [15] have also conducted an FTIR test to identify the common group in the hydrogel constituent. Figure 3 shows the FTIR test results obtained on the prepared ink of different compositions of sodium alginate and HA.

The result in Fig. 3 shows that hydrogen-bonded spectra (O–H stretching) occur at around 3400 cm^{-1} – 3600 cm^{-1} indicating the presence of a hydroxyl group for each composition (X axis-Wavenumber cm^{-1} , Y axis-Transmittance %).

The presence of the PO_4^{3-} group at the peak of 1000 – 1098 cm^{-1} indicates the presence of hydroxyapatite in the hydrogel.

The HA-related band shows the presence of the –OH group at a peak around 3000 – 3451 cm^{-1} for all compositions. Amide I($\text{C}=\text{O}$) i.e. the peak of primary amine around 1625 cm^{-1} , Amide II($\text{N}-\text{H}$ bending) at 1590 cm^{-1} gives the information for the protein content, Amide III ($\text{C}-\text{N}$ stretching) at 1279 – 1300 cm^{-1} indicates the presence of peptides bonds for the secondary structure of the protein, and A band of carbonate (CO_3^{2-}) at 1400 – 1500 indicates the presence of carbonate substitution for the formation of HA lattice. The C–H group around 2851 cm^{-1} – 2922 cm^{-1} indicates the presence of lipids and protein. The presence of a carboxyl group ($-\text{COO}-$ symmetric) peak at 1422 cm^{-1} indicates the presence of alginate. All the bands obtained from the FTIR test on prepared ink are suitable for bone regeneration. Maximum intensities in the graph of 6 wt% Alg and 24 wt% HA indicate that a scaffold prepared from this ink will be more suitable for bone formation. A similar kind of FTIR result was obtained by the researchers Adhikari et al. [15]; Basu et al. [22]; Hossan et al. [18]; Shaheen et al. [23].

SEM Analysis

A rough surface is more favourable for promoting cell adhesion and it is desirable in tissue engineering applications. SEM (scanning electron microscope) test helps in understanding the structural characteristics at the microscale level. SEM provide surface morphology, high-resolution images, and pore size distribution as done by different researchers like Basu et al. [24]; Iwanaga et al. [25]; and Turco et al. [26]. Figure 4 represents the XRD plot and SEM images of HA.

The XRD plot in Fig. 4a shows peaks at ≈ 25.9 , 31.77 , 32.9° , and 34.0° confirming the presence of crystal planes of hydroxyapatite [27]. Figure 4b shows the SEM images

Table 2 Relationship between shear rate, shear stress and viscosity

Point no	Pure Alg, 6 wt%			6 wt% Alg + 8 wt% HA			6 wt% Alg + 16 wt% HA			6 wt% Alg + 24 wt% HA		
	Shear rate	Shear stress	Viscosity	Shear rate	Shear stress	Viscosity	Shear rate	Shear stress	Viscosity	Shear rate	Shear stress	Viscosity
	[1/s]	[Pa]	[mPa·s]	[1/s]	[Pa]	[mPa·s]	[1/s]	[Pa]	[mPa·s]	[1/s]	[Pa]	[mPa·s]
1	0.0099	0.5431	54,392	0.00997	1.0885	1.09E+05	0.0099	1.7141	1.72E+05	0.00997	2.481	2.49E+05
2	0.0133	1.0526	78,984	0.0133	2.2711	1.70E+05	0.0133	3.6992	2.78E+05	0.0133	5.3968	4.05E+05
3	0.0178	1.5206	85,559	0.0178	3.425	1.93E+05	0.0178	5.658	3.18E+05	0.0178	8.2375	4.63E+05
4	0.0237	2.048	86,422	0.0237	4.7297	2.00E+05	0.0237	7.8806	3.33E+05	0.0237	11.4	4.81E+05
5	0.0316	2.686	84,995	0.0316	6.2679	1.98E+05	0.0316	10.472	3.31E+05	0.0316	15.039	4.76E+05
6	0.0421	3.4772	82,503	0.0421	8.1121	1.92E+05	0.0421	13.531	3.21E+05	0.0421	19.274	4.57E+05
7	0.0562	4.4691	79,522	0.0562	10.337	1.84E+05	0.0562	17.164	3.05E+05	0.0562	24.228	4.31E+05
8	0.0749	5.7263	76,411	0.0749	13.057	1.74E+05	0.0749	21.486	2.87E+05	0.0749	30.036	4.01E+05
9	0.0999	7.316	73,207	0.0999	16.381	1.64E+05	0.0999	26.605	2.66E+05	0.0999	36.821	3.68E+05
10	0.133	9.3167	69,865	0.133	20.378	1.53E+05	0.133	32.561	2.44E+05	0.133	44.653	3.35E+05
11	0.178	11.689	65,750	0.178	24.971	1.40E+05	0.178	39.209	2.20E+05	0.178	53.368	3.00E+05
12	0.237	14.462	60,987	0.237	30.032	1.27E+05	0.237	46.349	1.95E+05	0.237	62.63	2.64E+05
13	0.316	17.527	55,427	0.316	35.414	1.12E+05	0.316	53.926	1.71E+05	0.316	72.276	2.29E+05
14	0.422	20.851	49,449	0.422	41.166	97,625	0.422	61.905	1.47E+05	0.422	82.194	1.95E+05
15	0.562	24.498	43,565	0.562	47.508	84,483	0.562	70.524	1.25E+05	0.562	92.871	1.65E+05
16	0.75	28.579	38,111	0.75	54.62	72,837	0.75	80.111	1.07E+05	0.75	104.56	1.39E+05
17	1	33.338	33,340	1	62.754	62,755	1	90.865	90,865	1	117.46	1.17E+05
18	1.33	38.78	29,082	1.33	71.961	53,963	1.33	102.93	77,184	1.33	131.92	98,928
19	1.78	44.9	25,250	1.78	82.354	46,312	1.78	116.23	65,363	1.78	147.55	82,970
20	2.37	51.828	21,857	2.37	93.678	39,504	2.37	130.77	55,147	2.37	164.63	69,423
21	3.16	59.632	18,858	3.16	106.24	33,598	3.16	146.38	46,287	3.16	182.95	57,853
22	4.22	68.396	16,220	4.22	119.95	28,443	4.22	163.13	38,684	4.22	202.58	48,039
23	5.62	78.066	13,883	5.62	134.86	23,982	5.62	181.01	32,190	5.62	223.23	39,697
24	7.5	88.579	11,813	7.5	150.75	20,103	7.5	199.85	26,651	7.5	244.88	32,655
25	10	99.904	9991.4	10	167.81	16,781	10	219.65	21,966	10	267.26	26,726
26	13.3	112.37	8427	13.3	185.86	13,938	13.3	240.23	18,015	13.3	290.3	21,771
27	17.8	125.58	7062.4	17.8	204.64	11,508	17.8	261.11	14,684	17.8	313.69	17,642
28	23.7	139.87	5898.7	23.7	224.32	9460.1	23.7	282.7	11,922	23.7	337.53	14,235
29	31.6	154.96	4900.6	31.6	244.41	7730.1	31.6	304.54	9631.9	31.6	361.14	11,422
30	42.2	170.94	4053.8	42.2	264.66	6276.8	42.2	326.17	7735.9	42.2	384.33	9115.3
31	56.2	187.62	3336.5	56.2	285.03	5069.2	56.2	347.4	6178.7	56.2	406.97	7238.1
32	75	205.02	2734	75	305.32	4072	75	368.22	4911	75	428.36	5713.1
33	100	222.99	2229.8	100	325.22	3252.5	100	388.36	3884	100	448.89	4489.6
34	133	241.33	1809.6	133	345.23	2589	133	407.49	3056	133	468.47	3513.4

Table 2 (continued)

Point no	Pure Alg, 6 wt%			6 wt% Alg + 8 wt% HA			6 wt% Alg + 16 wt% HA			6 wt% Alg + 24 wt% HA		
	Shear rate	Shear stress	Viscosity	Shear rate	Shear stress	Viscosity	Shear Rate	Shear stress	Viscosity	Shear Rate	Shear stress	Viscosity
35	178	260.03	1462.1	178	364.94	2052.3	178	427.19	2402.4	178	487.65	2742.5
36	237	279.21	1177.4	237	385.27	1624.7	237	445.61	1879.2	237	506.94	2137.8
37	316	298.93	945.22	316	406.3	1284.8	316	465.71	1472.7	316	527.05	1666.7
38	422	319.32	757.12	422	428.07	1015.1	422	486.71	1154.1	422	547.89	1299.2
39	563	341.38	606.9	562	452.93	805.25	562	510.2	907.1	562	563.94	1002.7
40	750	365.73	487.53	750	479.44	639.15	750	534.33	712.33	750	528.63	704.7
41	1.00E+03	393.42	393.24	1.00E+03	498	497.8	1.00E+03	470.76	470.53	1.00E+03	407.17	406.96

of HA at different scales such as 10 μm , 20 μm , 100 μm , 500 μm , 2.00 mm.

From Fig. 5, it is observed that increasing the hydroxyapatite content results in increasing the surface roughness. Higher surface roughness provides the proper addition of layer by layer in the 3D printed scaffold identified. From the obtained result 6 wt% Alg + 24 wt% HA shows the higher surface roughness and is suitable for bone regeneration.

EDS Mapping

The implementation of an energy-dispersive X-ray Spectrometry (EDX) test in the context of bio-printed scaffolds for bone serves as a crucial analytical tool [28]. EDX enables the precise characterization of the scaffold's composition, particularly in terms of calcium (Ca) and phosphorus (P) content. In the intricate process of bone regeneration, the Ca/P ratio is of paramount significance. This ratio is a key indicator of the scaffold's biocompatibility and its ability to mimic the natural mineral composition of bone tissue. A well-balanced Ca/P ratio is essential for optimal bone mineralization and overall scaffold effectiveness. Therefore, the EDX test not only validates the presence of essential elements like calcium but also provides a quantitative measure through the Ca/P ratio, aiding in the fine-tuning and enhancement of bio-printed scaffolds for more accurate and biomimetic bone tissue engineering. All scaffolds were analysed by energy-dispersive X-ray spectrometry (EDX) to confirm the production of calcium [18]. EDX analysis confirmed the above results. Calcium and phosphate contents on the cell surface were found in all scaffolds as shown in Fig. 6.

The result in Fig. 6 shows that the Ca/P ratio is in the range of 1.12–1.68 Ca/P ratios for all the compositions and is lying in the range for bone formation. As increasing Ca/P ratios > 2 will decrease cell viability and Ca/P ratio less than 2 and 1 will enhance osteoblast viability. So the Ca/P ratio should be in the range of 1–2 as suggested by Liu et al. [29]. So, the result obtained from the EDS analysis 6 wt% Alg + 24 wt% HA is most suitable for bone formation.

TGA/DTA Testing

TGA measures the changes in the mass as a function of temperature which is very useful for determining the composition of the scaffold [30]. TGA provide information regarding the thermal stability of the scaffold. TGA helps in finding out the temperature profile on which the degradation occurs which helps to predict the scaffold life span and its compatibility with the natural bone regeneration process. DTA helps in identifying the phase transition such as melting point or crystallization which is helpful for understanding the thermal behaviour of the scaffold material.

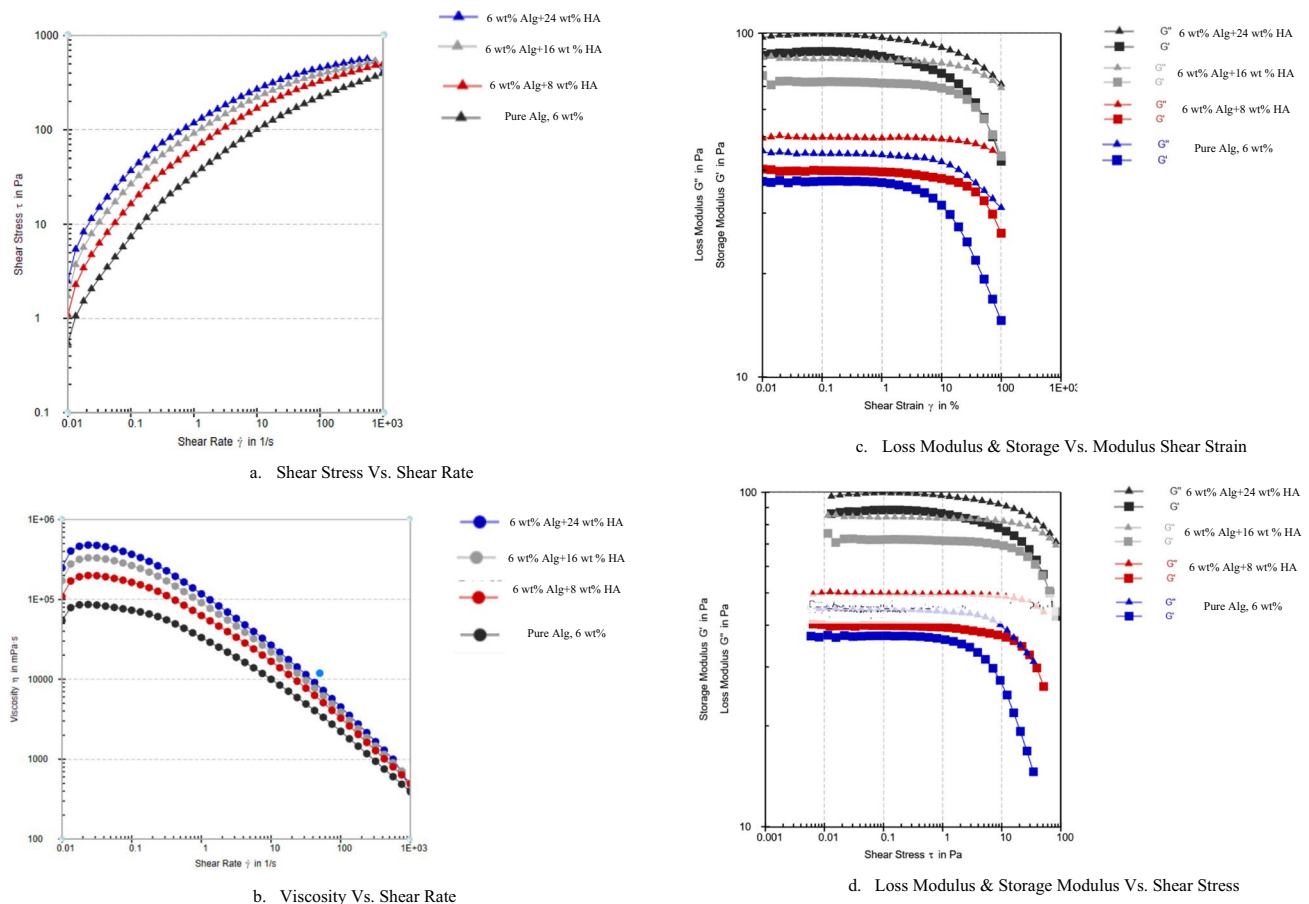


Fig. 2 Flow curve, **a** shear stress versus shear rate; **b** viscosity versus shear rate; **c** loss modulus and storage modulus versus shear strain; **d** loss modulus and storage modulus versus shear stress

A thermogravimetric analysis (TGA) and differential thermal analysis were conducted to examine the thermal stability and the crystalline condition of the alginate and hydroxyapatite composite material. Alginate and hydroxyapatite composite were heated from room temperature to 600 °C within increments in the temperature of 10 °C per minute in nitrogen as suggested by Hossan et al. [18].

From the TGA curve in Fig. 7, it can be noticed that 65.3% of the total mass of the pure alginate sample decomposed along with the loss of water at 340 °C and 4.4% of the total mass remains at 600 °C for pure Alginate. For 6 wt% Alg+8 wt% HA, 11.8 wt% of the total mass of the Alginate sample decomposed at 260 °C and 9.5% of the total mass remains at 600 °C. For 6 wt% Alg+ 16 wt% HA, 10.4% of the total mass of Alginate is decomposed at 260 °C and 5.4% of the total mass remains at 600 °C. For 6 wt% Alg+24 wt% HA, 20% of the total mass of the alginate sample remains at 260 °C and 4.7% of the total mass remains at 600 °C. Ruptures in the chains and monomers of the alginate are the cause of the mass loss in the sample [18, 19].

In the DTA curve, the endothermic peaks are at ≈ 70 °C and the exothermic peaks are at 312 °C for pure alginate. For 6 wt% Alg+8 wt% HA endothermic reactions starts ≈ 55 °C and the exothermic peak is at 296.4 °C. For 6 wt% Alg+16 wt% HA endothermic peak is at ≈ 75 °C and the exothermic peak is at 311.6 °C. For 6 wt%+24 wt% HA endothermic peak is at ≈ 85 °C and 318.7 °C. The endothermic peak in the DTA curve is due to the vaporization and absorption of water and alginate molecules. The exothermic curve is due to oxidation and polymerization of alginate and hydroxyapatite particles. By increasing the concentration of HA, the total degradation rate is reduced as shown in the TGA graph. In DTA curves, the endothermic peak and exothermic peak temperature increases by increasing the concentration of HA. The exothermic peak value is maximum for 6 wt%+24 wt% HA. Including HA composite in alginate provides more thermal stability in the composite material.

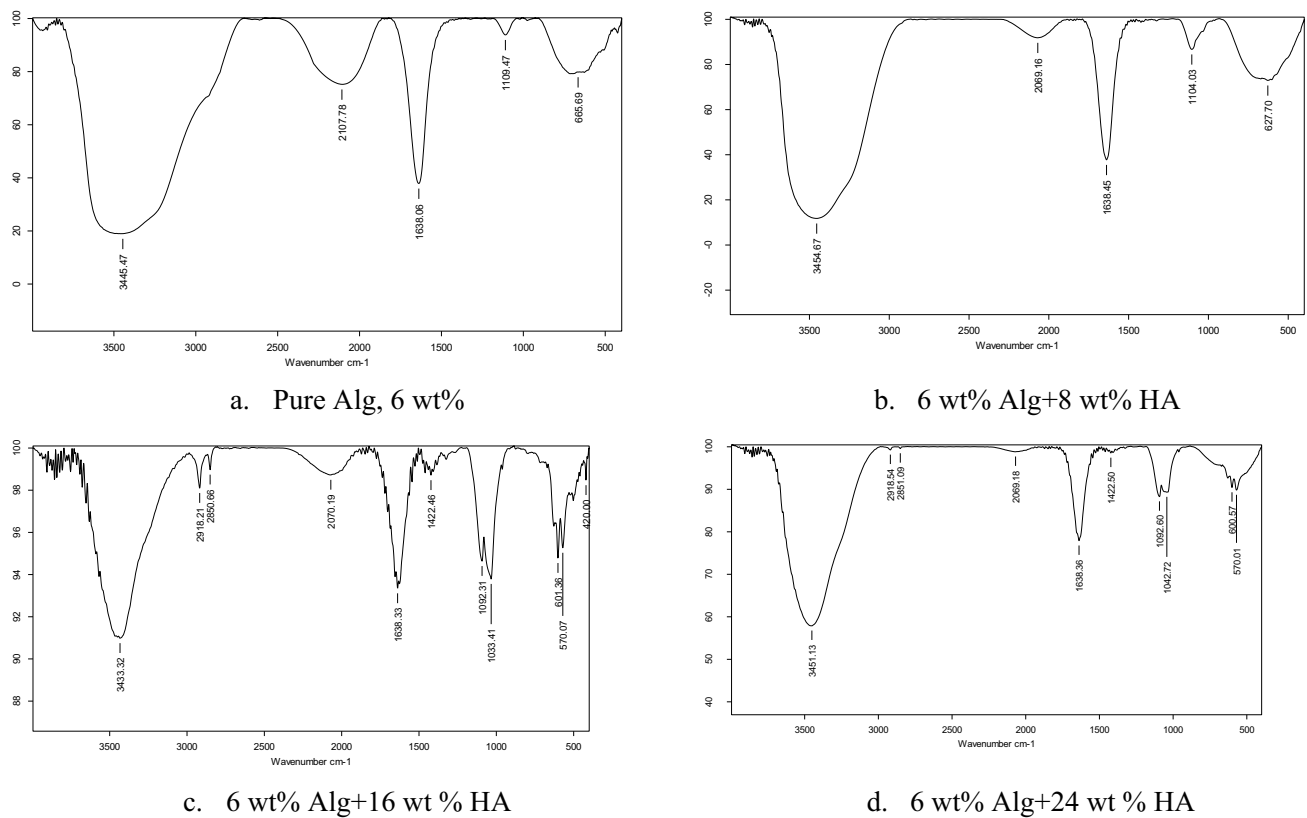


Fig. 3 FTIR analysis of biomaterial inks

Conclusion

In this study, the Trivima basic bioprinter was used to successfully print the scaffold composed of alginate and hydroxyapatite-based biomaterials ink. CaCl_2 was used as a crosslinker to obtain a stable structure after printing. The rheology testing confirms the shear stress range of prepared ink is suitable for extrusion-based bioprinters. Viscosity reduction under shear was observed in all types of biomaterial ink combinations used in this study. Extrusion pressure increases by increasing HA content. FTIR study confirms the interaction between Alginate and Hydroxyapatite. The presence of different component bands indicates that the prepared biomaterial inks

are suitable for bone tissue regeneration. Surface roughness was observed by using the SEM images. The surface morphology observed through SEM images indicated the potential of prepared biomaterial inks for cell growth and tissue regeneration. EDS analysis confirmed the obtained Ca/P ratio range of the suitability of formed biomaterial-based ink suitable for Bone application. TGA/DTA provides the thermal stability of Alginate and HA biomaterial ink with degradation phases. Future studies should focus on compressive testing on the bioprinted bone scaffold. Future researchers should add Synthetic biomaterials within the Alginate and hydroxyapatite-based biomaterial ink to enhance the mechanical strength.

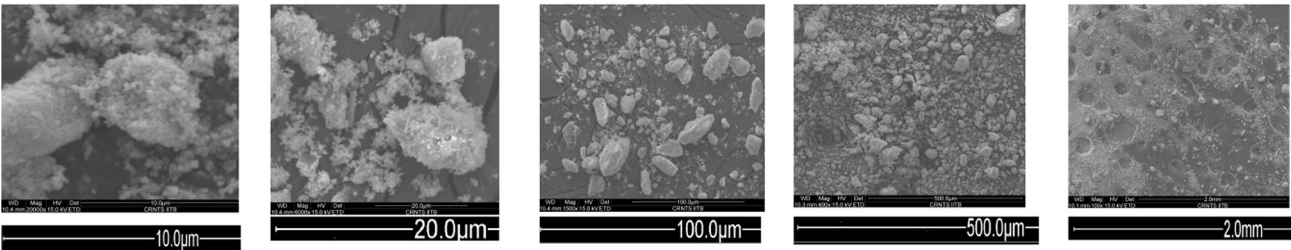
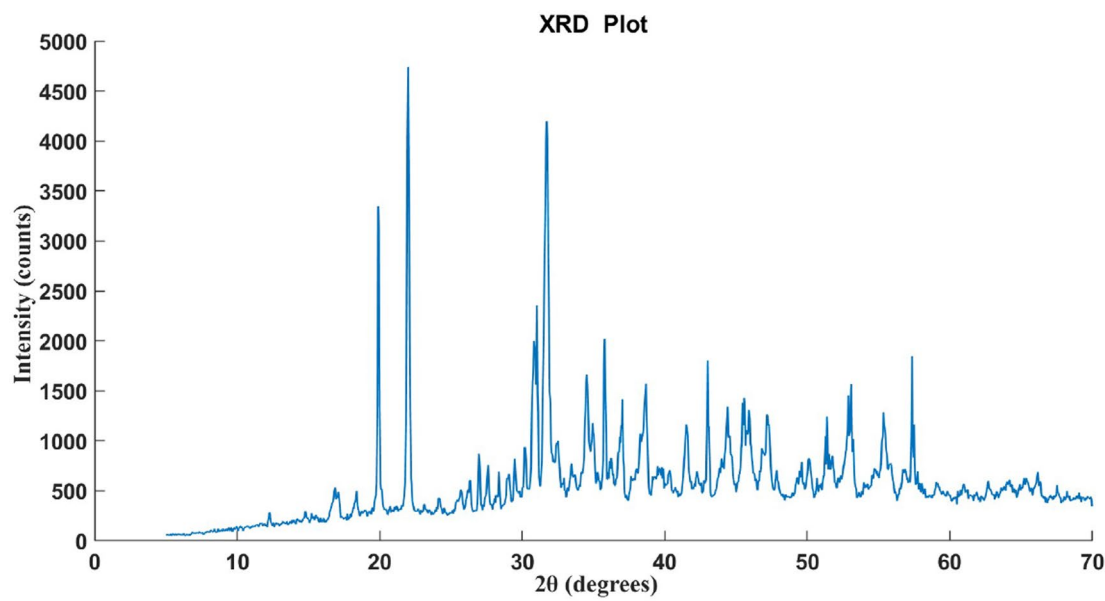


Fig. 4 **a** XRD plot of HA; **b** SEM images of HA

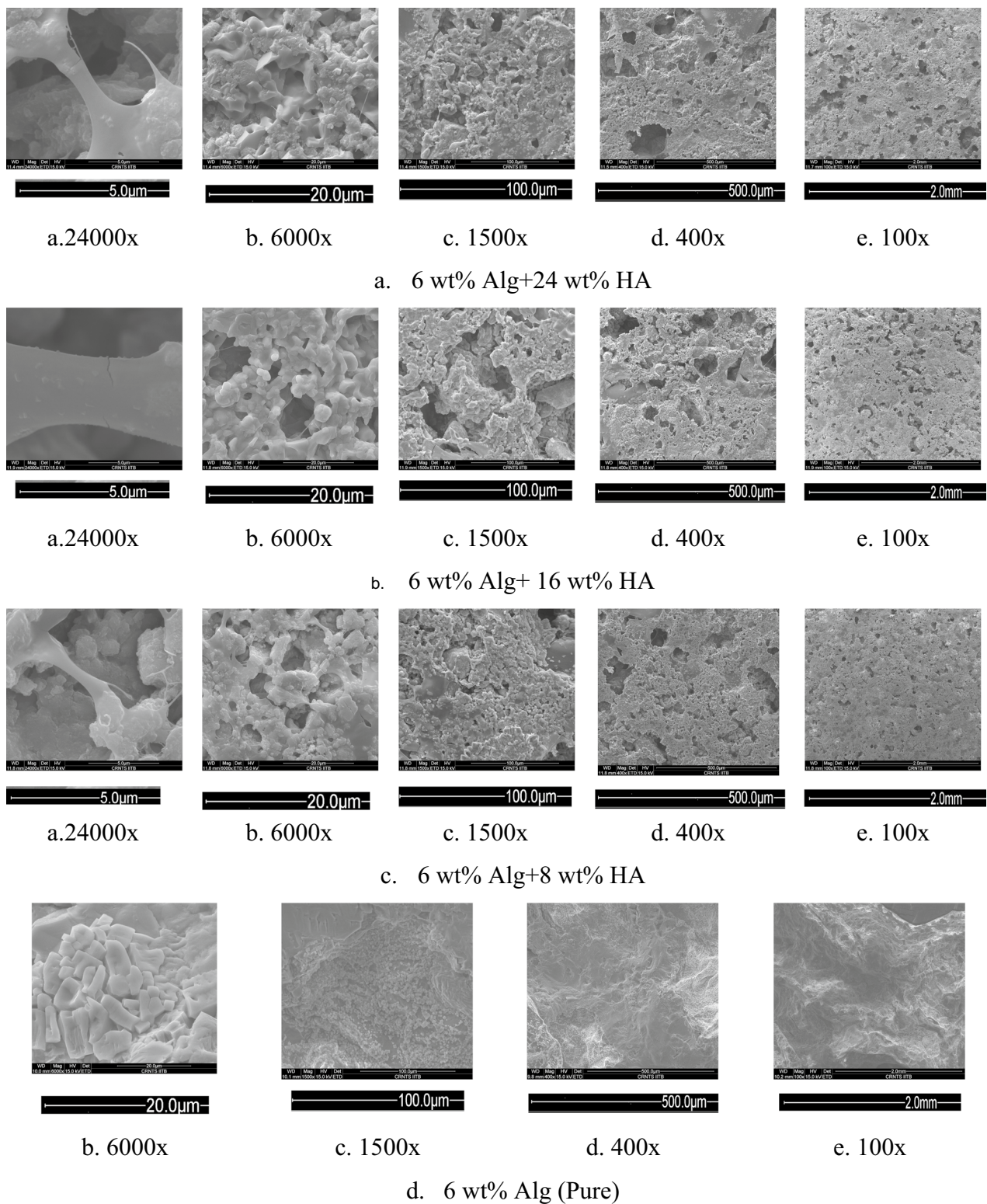


Fig. 5 SEM image of microstructure of biomaterial ink of different concentrations here magnification is denoted as x

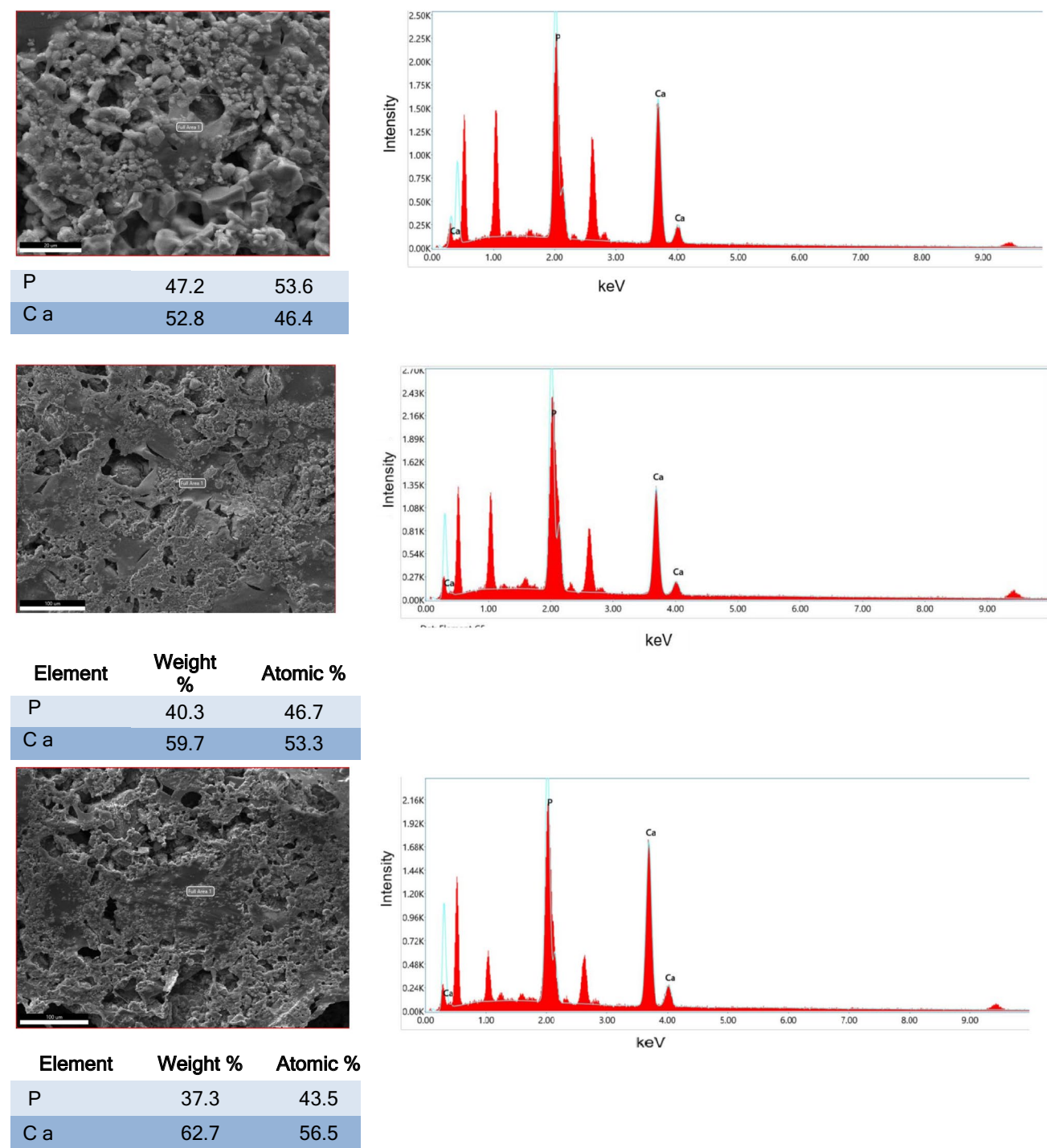
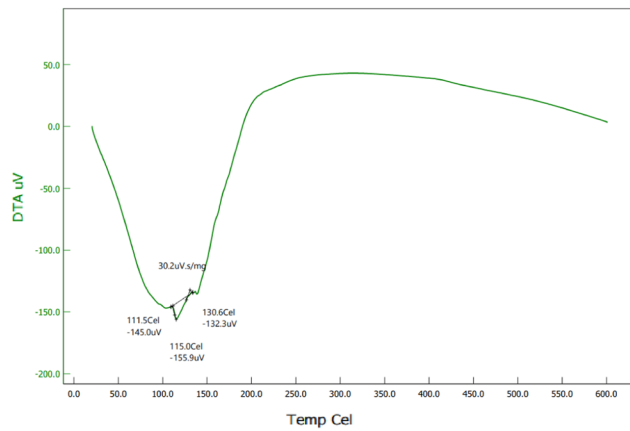
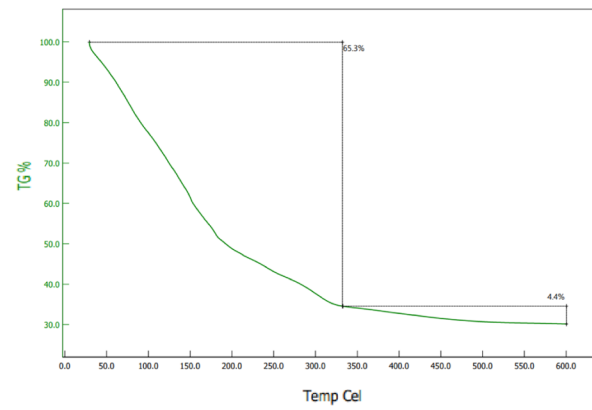


Fig. 6 EDX analysis of bioprinted scaffold

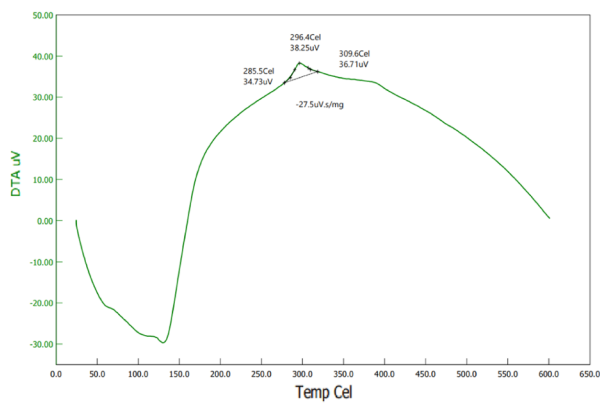


1.DTA curve

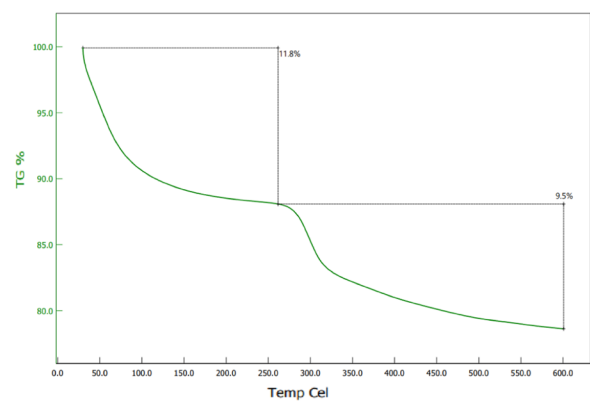


2.TGA curve

(a) Pure Alginate

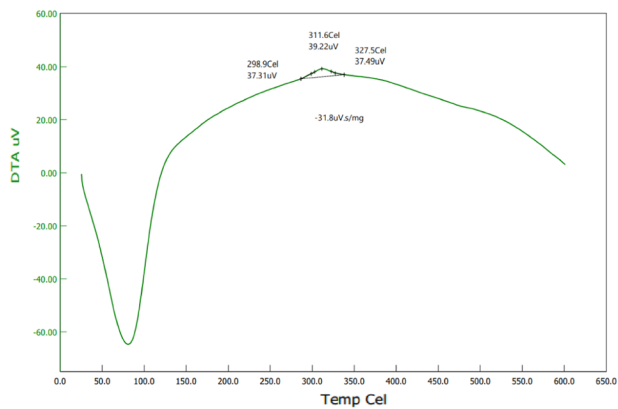


1. DTA curve

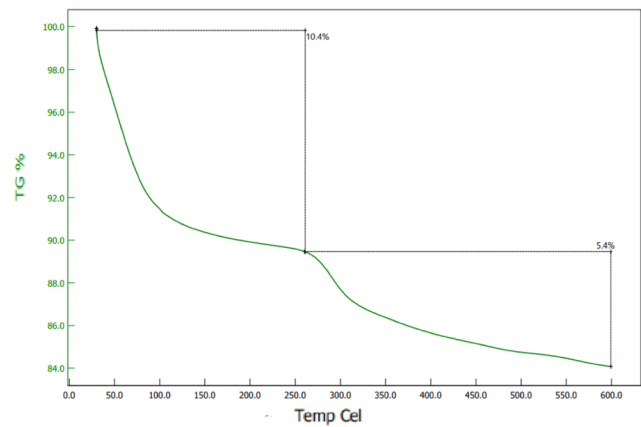


2. TGA curve

(b) 6 wt% Alg+8 wt% HA



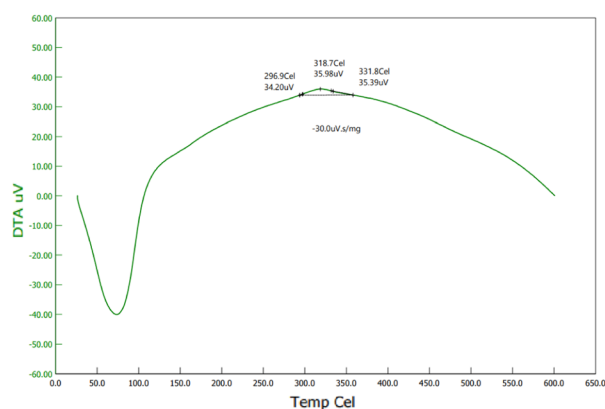
1. DTA curve



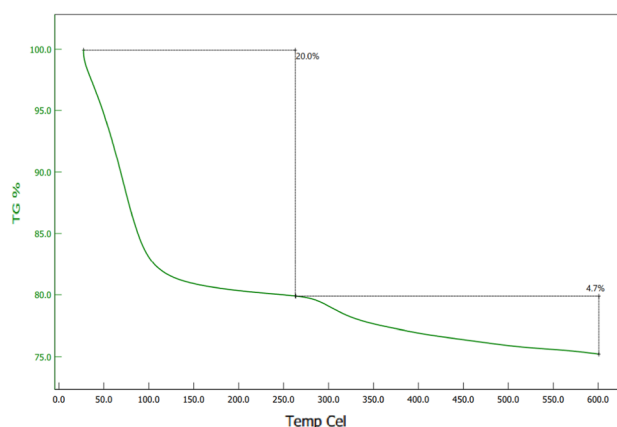
2.TGA curve

(c) 6 wt% Alg+16 wt% HA

Fig. 7 DTA and TGA analysis



1. DTA curve



2. TGA curve

(d) 6 wt% Alg+24 wt% HA

Fig. 7 (continued)

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Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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